Differences in niche similarity between groups

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LOAD PAKAGES

```
library(dplyr)
library(tidyverse)
library(kableExtra)
library(ggplot2)
library(car)
library(readxl)
library(dunn.test)
```

```
cat("Versions of the libraries used:\n")
```

```
## Versions of the libraries used:
```

```
cat("dplyr:", as.character(packageVersion("dplyr")), "\n") # dplyr: 1.1.4
```

```
## dplyr: 1.1.4
```

```
cat("tidyverse:", as.character(packageVersion("tidyverse")), "\n") # tidyverse: 2.0.0
```

```
## tidyverse: 2.0.0
```

```
cat("kableExtra:", as.character(packageVersion("kableExtra")), "\n") # kableExtra: 1.4.0
```

```
## kableExtra: 1.4.0
```

```r
cat("ggplot2:", as.character(packageVersion("ggplot2")), "\n") # ggplot2:4.0.0
```

```
## ggplot2: 4.0.0
```

```r
cat("car:", as.character(packageVersion("car")), "\n") # car: 3.1.3
```

```
## car: 3.1.3
```

```r
cat("readxl:", as.character(packageVersion("readxl")), "\n") # readxl: 1.4.5
```

```
## readxl: 1.4.5
```

```r
cat("dunn.test:", as.character(packageVersion("dunn.test")), "\n") # dunn.test: 1.3.6
```

```
## dunn.test: 1.3.6
```

LOAD DATA

```r
all_sp <- read.csv("02_Hypervolume/R_Overlap_pairs_wise/results_overlap_all_species.csv")
lentic_sp <- read.csv("02_Hypervolume/R_Overlap_pairs_wise/results_overlap_Lentic.csv")
lotic_sp <- read.csv("02_Hypervolume/R_Overlap_pairs_wise/results_overlap_Lotic.csv")
```

Select lentic vs. lotic species pairs and lotic vs. lentic species pairs

```r
species_sp1 = unique(all_sp$sp1)
```

```r
species_sp2 = unique(all_sp$sp2)
```

```r
sp1_df = as.data.frame(species_sp1)
sp2_df = as.data.frame(species_sp2)

#write.csv(sp1_df, "sp1_df.csv")
#write.csv(sp2_df, "sp2_df.csv")
```

```r
species_group <- read_excel("01_Data/03_Data_species/List_of_species.xlsx", sheet = "species_group")
```

```r
head(species_group)
```

```
## # A tibble: 6 x 2
##   species group
##   <chr>   <chr>
## 1 sp_1    Lentic
## 2 sp_10   Lentic
## 3 sp_100  Lentic
## 4 sp_101  Lentic
## 5 sp_102  Lentic
## 6 sp_103  Lentic
```

```r
head(all_sp)
```

```
##    sp1    sp2      jaccard    sorensen frac_unique_1 frac_unique_2   distance
## 1 sp_1  sp_10 0.2738750215 0.429987270    0.68215485    0.33559864 0.32791462
## 2 sp_1 sp_100 0.7684291663 0.869052808    0.20063389    0.04794989 0.05769258
## 3 sp_1 sp_101 0.4561846367 0.626547795    0.52178624    0.09171813 0.18119259
## 4 sp_1 sp_102 0.8900669579 0.941836430    0.07222737    0.04366683 0.01560695
## 5 sp_1 sp_103 0.5628540560 0.720289977    0.05112961    0.41954107 0.22452750
## 6 sp_1 sp_104 0.0006053584 0.001209984    0.99939464    0.00000000 0.86233539
```

```r
# First, we group together each species.
all_sp_grouped <- all_sp %>%
  # We add the group information for sp1
  left_join(species_group, by = c("sp1" = "species")) %>%
  rename(group_sp1 = group) %>%
  # We add the group information for sp2
  left_join(species_group, by = c("sp2" = "species")) %>%
  rename(group_sp2 = group)
```

```r
# Now we filter: we only keep pairs of species from different groups.
all_sp_filtered <- all_sp_grouped %>%
  filter(group_sp1 != group_sp2)

# result
head(all_sp_filtered)
```

```
##    sp1    sp2     jaccard     sorensen frac_unique_1 frac_unique_2  distance
## 1 sp_1 sp_108 0.001357738 0.0027117943     0.9986416    0.26315789 0.5937005
## 2 sp_1 sp_110 0.023238862 0.0454221644     0.9767255    0.06177606 0.4751361
## 3 sp_1 sp_120 0.252539944 0.4032445350     0.7443931    0.04535714 0.2012827
## 4 sp_1 sp_121 0.000299220 0.0005982609     0.9997008    0.00000000 0.9170939
## 5 sp_1 sp_122 0.221276567 0.3623693000     0.7756123    0.05896510 0.2148859
## 6 sp_1  sp_16 0.456080625 0.6264496860     0.5323189    0.05158038 0.1189897
##   group_sp1 group_sp2
## 1    Lentic     Lotic
## 2    Lentic     Lotic
## 3    Lentic     Lotic
## 4    Lentic     Lotic
## 5    Lentic     Lotic
## 6    Lentic     Lotic
```

```r
#write.csv(all_sp_filtered, "03_Overlap_pairs_wise/results_overlap_LenticVSLotic_species_v1.csv")
```

Analysis

```r
nrow(all_sp_filtered)
```

```
## [1] 2940
```

```r
nrow(lentic_sp)
```

```
## [1] 3486
```

```r
nrow(lotic_sp)
```

```
## [1] 595
```

```r
head(all_sp_filtered)
```

```
##    sp1    sp2     jaccard     sorensen frac_unique_1 frac_unique_2  distance
## 1 sp_1 sp_108 0.001357738 0.0027117943     0.9986416    0.26315789 0.5937005
## 2 sp_1 sp_110 0.023238862 0.0454221644     0.9767255    0.06177606 0.4751361
## 3 sp_1 sp_120 0.252539944 0.4032445350     0.7443931    0.04535714 0.2012827
## 4 sp_1 sp_121 0.000299220 0.0005982609     0.9997008    0.00000000 0.9170939
## 5 sp_1 sp_122 0.221276567 0.3623693000     0.7756123    0.05896510 0.2148859
## 6 sp_1  sp_16 0.456080625 0.6264496860     0.5323189    0.05158038 0.1189897
##   group_sp1 group_sp2
## 1    Lentic     Lotic
## 2    Lentic     Lotic
## 3    Lentic     Lotic
## 4    Lentic     Lotic
## 5    Lentic     Lotic
## 6    Lentic     Lotic
```

```r
head(lentic_sp)
```

```
##   X  sp1    sp2      jaccard    sorensen frac_unique_1 frac_unique_2   distance
## 1 1 sp_1  sp_10 0.2752409470 0.431668929    0.68091178    0.33300020 0.32791462
## 2 2 sp_1 sp_100 0.7646979731 0.866661587    0.20283337    0.05056948 0.05769258
## 3 3 sp_1 sp_101 0.4547249577 0.625169666    0.52283810    0.09371595 0.18119259
## 4 4 sp_1 sp_102 0.8902403697 0.941933506    0.07213174    0.04356826 0.01560695
## 5 5 sp_1 sp_103 0.5601030838 0.718033429    0.05410226    0.42135955 0.22452750
## 6 6 sp_1 sp_104 0.0006053584 0.001209984    0.99939464    0.00000000 0.86233539
```

```r
head(lotic_sp)
```

```
##      sp1    sp2     jaccard    sorensen frac_unique_1 frac_unique_2  distance
## 1 sp_108 sp_110 0.067822048 0.127028747    0.08182601     0.9317655 0.5153931
## 2 sp_108 sp_120 0.001530875 0.003057069    0.77647059     0.9984609 0.5798189
## 3 sp_108 sp_121 0.000000000 0.000000000    1.00000000     1.0000000 0.9124705
## 4 sp_108 sp_122 0.001529120 0.003053571    0.80099502     0.9984614 0.5290165
## 5 sp_108  sp_16 0.001663136 0.003320748    0.55421687     0.9983334 0.6004856
## 6 sp_108  sp_18 0.016127697 0.031743445    0.06791569     0.9838533 0.4761823
```

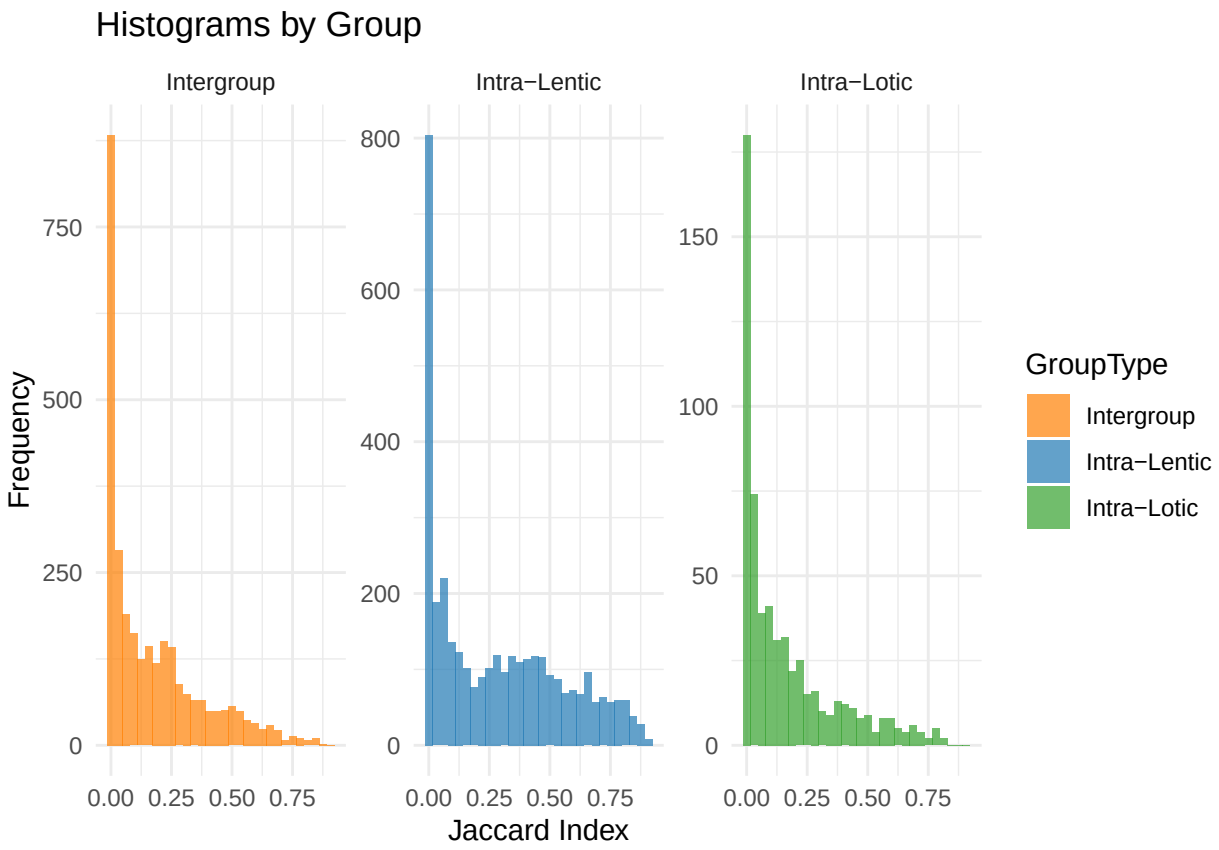
```r
# Create three dataframes (one for each type of comparison)
lentico_df <- data.frame(Jaccard = lentic_sp$jaccard, GroupType = "Intra-Lentic")
lotico_df <- data.frame(Jaccard = lotic_sp$jaccard, GroupType = "Intra-Lotic")
lentico_lotico_df <- data.frame(Jaccard = all_sp_filtered$jaccard, GroupType = "Intergroup")

# Combine all dataframes into one
combined_df <- rbind(lentico_df, lotico_df, lentico_lotico_df)
```

```r
#head(combined_df)
```

```r
colors = c("#ff7f00", "#1f78b4", "#33a02c")

ggplot(combined_df, aes(x = Jaccard, fill = GroupType)) +
    geom_histogram(position = "dodge", bins = 30, alpha = 0.7) +
    facet_wrap(~GroupType, scales = "free_y") + # Un gráfico por grupo
    scale_fill_manual(values = colors) + # <- Aquí aplicas tus colores
    labs(title = "Histograms by Group", x = "Jaccard Index", y = "Frequency") +
    theme_minimal()
```

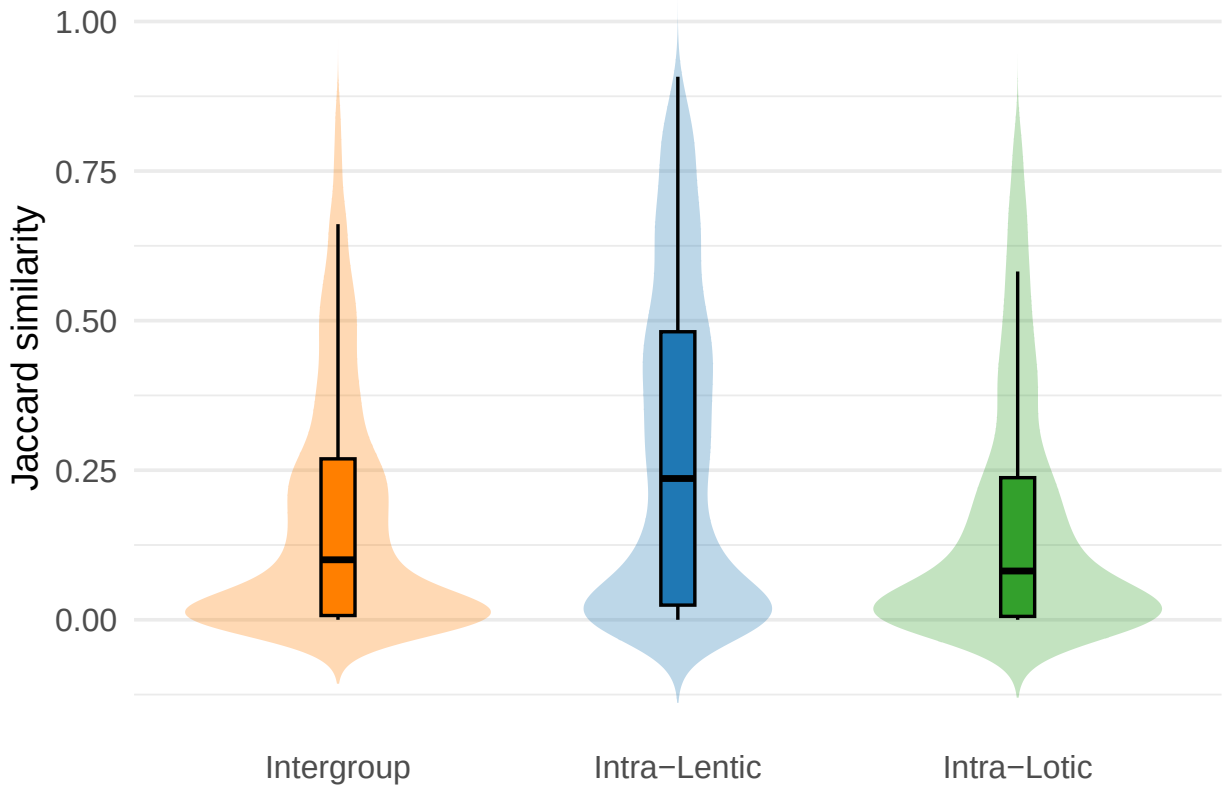



```r
# Kruskal-Wallis
kruskal.test(Jaccard ~ GroupType, data = combined_df)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  Jaccard by GroupType
## Kruskal-Wallis chi-squared = 275.75, df = 2, p-value < 2.2e-16
```

```r
boxplot <- ggplot(combined_df, aes(x = GroupType, y = Jaccard, fill = GroupType)) +
  geom_violin(trim = FALSE, alpha = 0.3, color = NA) +
  geom_boxplot(width = 0.1, color = "black", aes(fill = GroupType), outlier.shape = NA, linewidth = 0.6) +
  scale_fill_manual(values = colors) +
  scale_x_discrete(
    labels = c("Lentic_Lotic" = "Lentic vs Lotic",
```

```
                "Lentic_Lentic" = "Lentic vs Lentic",
                "Lotic_Lotic" = "Lotic vs Lotic")
  ) +
  labs(
    y = "Jaccard similarity",
    x = NULL
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
    axis.title.y = element_text(size = 14),
    axis.text = element_text(size = 12),
    panel.grid.major.x = element_blank(),
    panel.grid.minor.x = element_blank()
  )
boxplot
```




```
#ggsave("Figure_1.png", plot = boxplot, width = 5, height = 4, units = "in", dpi = 600)
```

```
# Post-hoc
library(dunn.test)
dunn.test(combined_df$Jaccard, combined_df$GroupType, kw = TRUE)
```

```
##   Kruskal-Wallis rank sum test
##
## data: x and group
```

```
## Kruskal-Wallis chi-squared = 275.7477, df = 2, p-value = 0
##
##
##                             Comparison of x by group
##                                  (No adjustment)
## Col Mean-|
## Row Mean |   Intergro   Intra-Le
## ---------+----------------------
## Intra-Le |  -15.46509
##          |    0.0000*
##          |
## Intra-Lo |   1.092146   9.837037
##          |     0.1374    0.0000*
##
## alpha = 0.05
## Reject Ho if p <= alpha/2
```